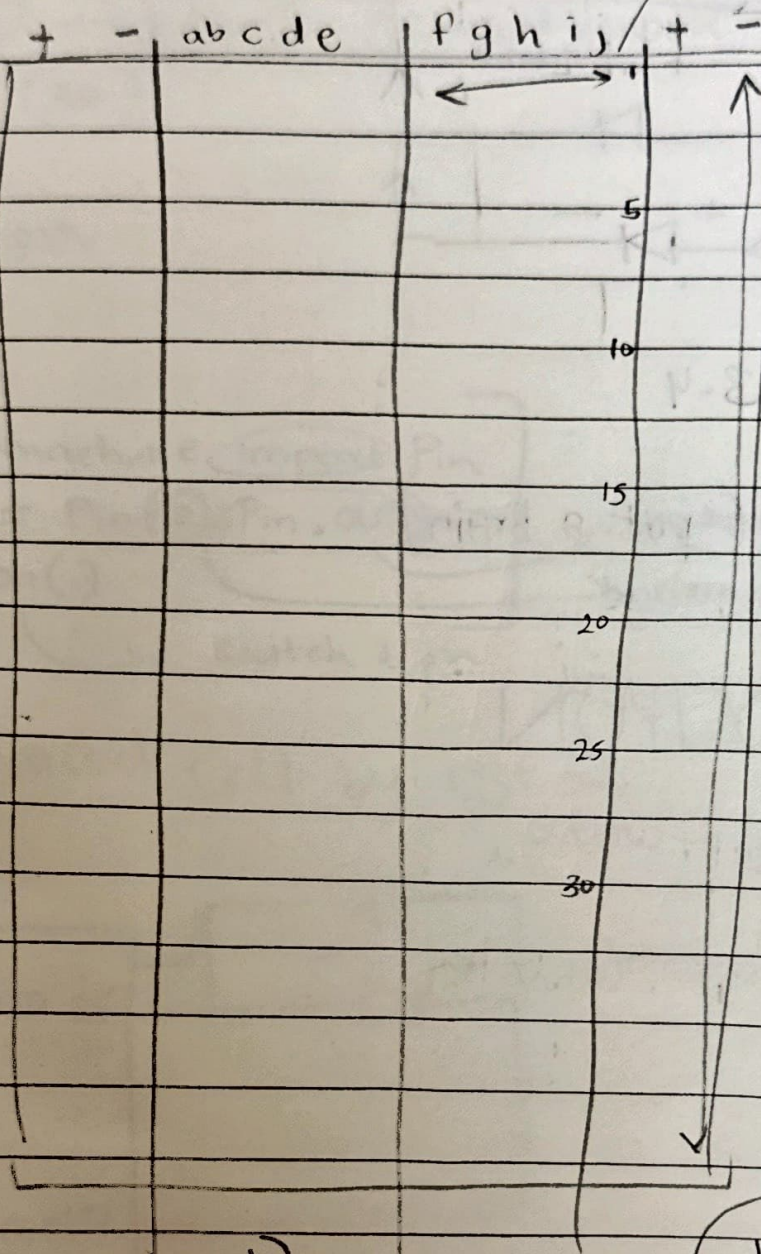


BREADBOARD :

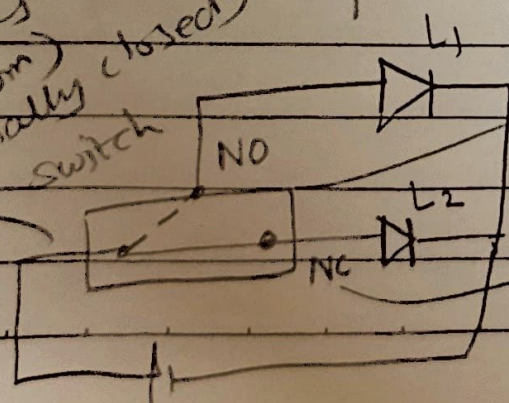


row wise connected

column wise connected

3 terminals
common (com)
NC (normally closed)

NO (normally open)

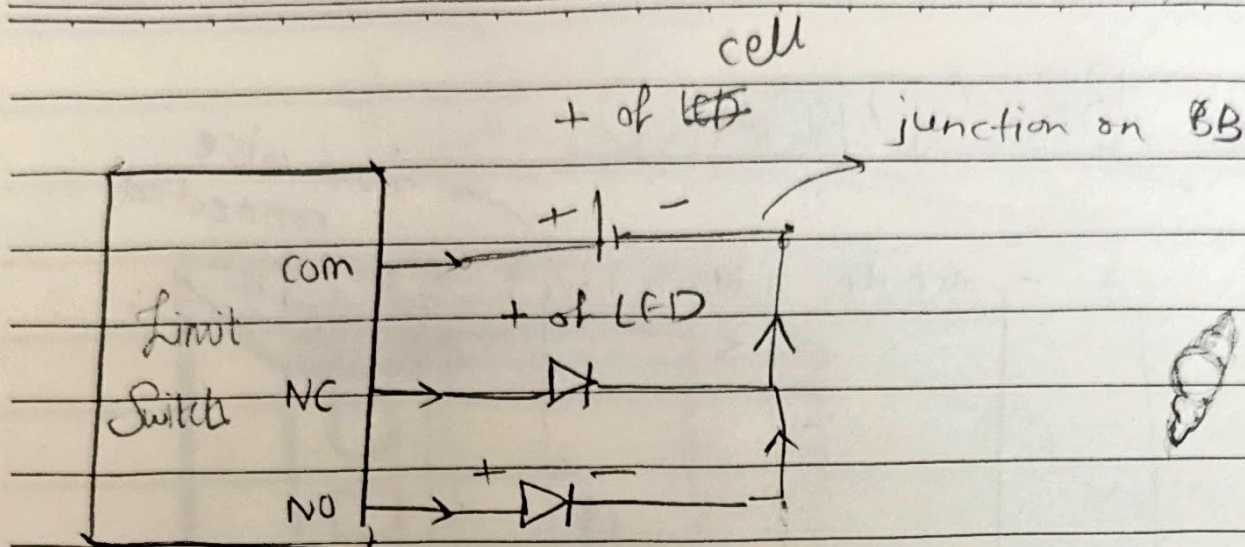


Limit Switch

2 LEDs

Black on switch
Blue (on)

where you want it to be always on



$|s \rightarrow 3.4$

To automate (put a brain) :

→ programming language

MICROPYTHON

hardware + software

ESP32 → microcontroller

Thorny

Shell Window

PIN CONF.

① GPIO → ~~output~~ general purpose input output
(30)

LCD

Sensor

```

from machine import Pin
astro = Pin(2, Pin.OUT)
astro.on()
    
```

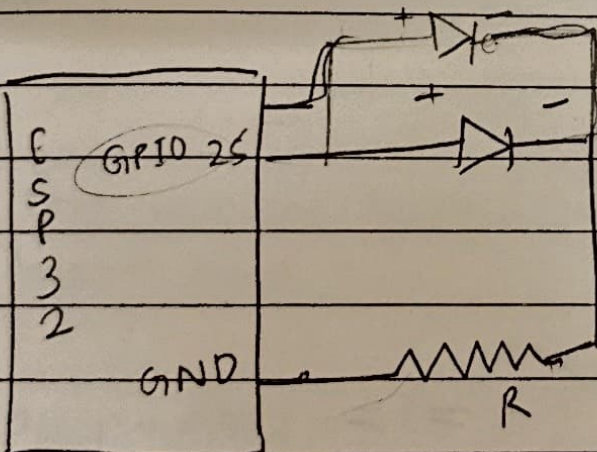
→ output
→ no. of switch
→ switch it on

Replaced cell by ESP32

GND → Ground
(- terminal)

+ resistor

value?



330 → more glow

510 → less glow

off → kuch nhi hota

DELAY

from machine import Pin

import time

astro = Pin(2, Pin.OUT)

astro.on()

time.sleep(2)

astro.off()

time.sleep(2)

Put time delay

repeat again

Count = 4

while (count < 3)

instructions

Reduce by 1

= sure

assigning

= =

checking

FINAL CODE

```
from machine import Pin
import time
```

```
astro = Pin(25, Pin.OUT) → creating an object  
astro
```

```
count = 5
```

```
while count > 3 :
```

```
    astro.on()
```

```
    time.sleep(2)
```

```
    astro.off()
```

```
    time.sleep(2)
```

```
    count = count - 1
```

~~#~~ for infinite times → while True

```
from machine import Pin
import time
```

```
astro = Pin(2, Pin.OUT)
```

```
bv = Pin(15, Pin.OUT)
```

NO

Date

ON

while True:

astro.on()

time.sleep(2)

astro.off()

~~bv.on()~~ time.sleep(2)

bv.on()

t.s(2)

bv.off()

t.s(2)